



## Self Attention Mechanism

In NLP, the main task was to convert words  $\rightarrow$  numbers  $\rightarrow$  vectorization

i) OHE (ONE HOT ENCODING)

|                | mat | cat | rat |
|----------------|-----|-----|-----|
| 1) mat cat rat | 1   | 0   | 0   |
| 2) cat rat rat | 0   | 1   | 0   |
|                | 0   | 0   | 1   |

num  $\rightarrow$  i)  $[1\ 0\ 0]$   $[0\ 1\ 0]$   $[1\ 0\ 0]$

$\rightarrow$  ii)  $[0\ 1\ 0]$   $[0\ 0\ 1]$   $[0\ 0\ 1]$

ii) Bag of words (BoW)

|       | mat | cat | rat |
|-------|-----|-----|-----|
| $s_1$ | 2   | 0   | 1   |
| $s_2$ | 0   | 2   | 1   |

$\rightarrow$  It creates a vector based on no. of times a word appear in the sentence

iii) TFidf and so on .....

These all techniques were good, but it fails to capture semantic meanings of the words.  
eg King and Queen are as different as King and hammer.

Word embeddings were introduced to capture semantic meaning (Embedding vectors) using word2vec technique such as GloVe, etc.

• In word embedding :-

- i) There could be 256, 512, 64 dimension vector
- ii) Each dimension represents a degree of "quality"
- iii) And based on training of those weights using cosine similarity, we update those weight vectors.

Similar  $\left[ \begin{array}{l} \text{King} \rightarrow [0.6 \quad 0.8 \quad 1 \quad 0 \quad 0.9] \\ \text{Queen} \rightarrow [0.7 \quad 0.75 \quad 0.80 \quad 0 \quad 0.85] \end{array} \right. > \text{Similar}$

- Maybe 1<sup>st</sup> dim i.e (0.6 & 0.7) could represent Royalty, 2<sup>nd</sup> dim i.e (0.8 & 0.75) could represent Power, Bravery and so on
- Since Queen & King are similar words their word embedding vectors should also look similar

\* Problems in word embedding :-

The problem of average meaning

eg:-

- 1) An apple a day keeps doctor away
- 2) Apple is healthy
- 3) Apple make great phones

word Embedding of apple  $\Rightarrow [x \quad y] \rightarrow [0.8 \quad 0.3]$   
 taste technology

(Based on above 3 datas)

- Based on above 3 datas, it has learned about "apple" more as fruit
- Now, if you provide apple as a company based input it would still treat apple as a fruit.
- word embedding create static embedding



The main problem for word embedding vector was it cannot capture ~~context~~ <sup>context</sup> based meaning of a word.

~~Self~~ Self-Attention solves the problem of context based meaning of a word.

### \* Self-Attention

S1: Money bank grows      S2: River bank flows

bank → bank

bank → नदी किनारा

• bank itself can be represented as:-

$$\text{bank} \rightarrow 0.3 \text{ money} + 0.7 \text{ bank} + 0.1 \text{ grows}$$

$$\text{bank} \rightarrow 0.5 \text{ river} + 0.3 \text{ bank} + 0.2 \text{ flows}$$

• Since contextual meaning of a sentence can be interpreted more by looking other words of a sentence

S<sub>1</sub>

S<sub>2</sub>

$$\text{money} = 0.7 \text{ money} + 0.2 \text{ bank} + 0.1 \text{ grows}$$

$$\text{river} = 0.8 \text{ river} + 0.15 \text{ bank} + 0.05 \text{ flows}$$

$$\text{bank} = 0.5 \text{ money} + 0.7 \text{ bank} + 0.05 \text{ grows}$$

$$\text{bank} = 0.2 \text{ river} + 0.78 \text{ bank} + 0.02 \text{ flows}$$

$$\text{grows} = 0.1 \text{ money} + 0.2 \text{ bank} + 0.7 \text{ grows}$$

$$\text{flows} = 0.4 \text{ river} + 0.01 \text{ bank} + 0.59 \text{ flows}$$

\* Therefore, the new embedding vector for  $s_1$  looks like

$$e_{\text{money}}^{(\text{new})} = 0.7 e_{\text{money}} + 0.2 e_{\text{bank}} + 0.1 e_{\text{grows}}$$

$$e_{\text{bank}}^{(\text{new})} = 0.25 e_{\text{money}} + 0.7 e_{\text{bank}} + 0.05 e_{\text{grows}}$$

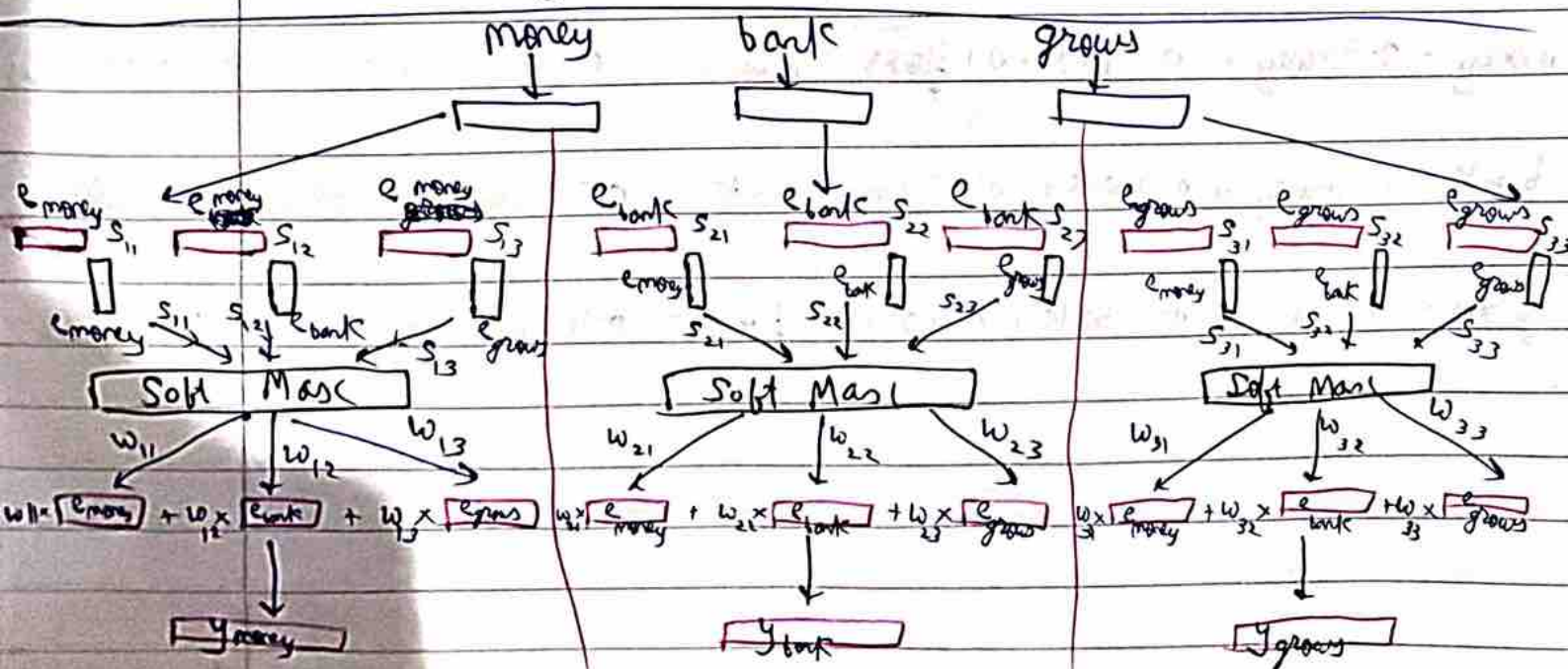
$$e_{\text{grows}}^{(\text{new})} = 0.1 e_{\text{money}} + 0.2 e_{\text{bank}} + 0.7 e_{\text{grows}}$$

\* but, what does the magnitude  $(0.7 \ 0.2 \ 0.1)$  mean?

- It's just degree of similarity of a word with other word and itself
- So, how does the magnitude arises, by taking dot product, i.e

$$e_{\text{bank}}^{(\text{new})} = \underbrace{[e_{\text{bank}} \cdot e_{\text{money}}^T]}_{0.25} e_{\text{money}} + \underbrace{[e_{\text{bank}} \cdot e_{\text{bank}}^T]}_{0.7} e_{\text{bank}} + \underbrace{[e_{\text{bank}} \cdot e_{\text{grows}}^T]}_{0.05} e_{\text{grows}}$$

$$e_{\text{bank}}^{(\text{new})} = 0.25 e_{\text{money}} + 0.7 e_{\text{bank}} + 0.05 e_{\text{grows}}$$

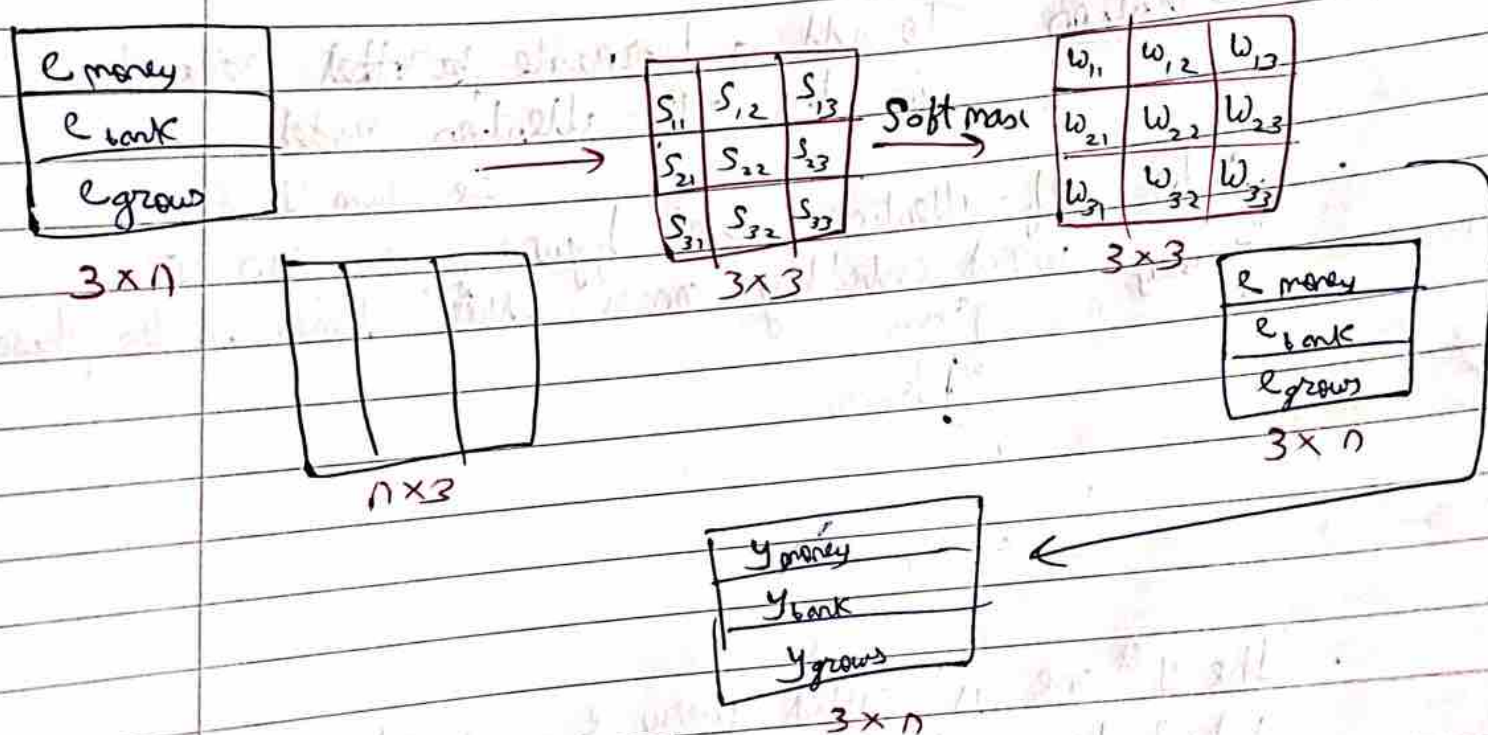


Self-attention model



So, all this operations in each column (i.e. 3 columns) can be done simultaneously, we can do parallelly. This solved the problem of training huge data, in previous Encoder-decoder archi. for LSTM, the output depend on previous input, so we can't do parallel operations which increase our training.

### \* Vector representation



### Improvements:-

- Parallel operations can be done

### Problem

- No parameter is involved i.e. learnable parameter

The problem was that it could not capture task ~~specific~~ specific contextual embedding

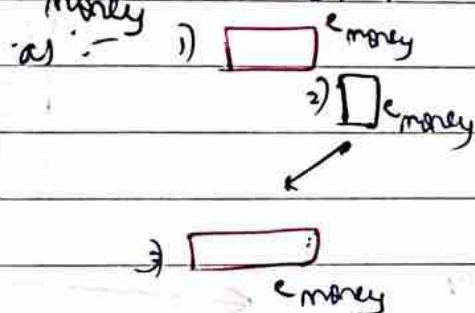
eg:- price of cake <sup>translate</sup> केक का दूकड़ा (wrong)  
price of cake is an idiom means बहुत आसान काम <sup>translation</sup>

So, since there are no parameters to learn, it could not capture task specific contextual embedding

Solution :- To add a learnable ~~parameter~~ parameter in the self-attention model.

see column-1 (first)

In self-attention model, figure, you can see word embedding comes three times in the phases

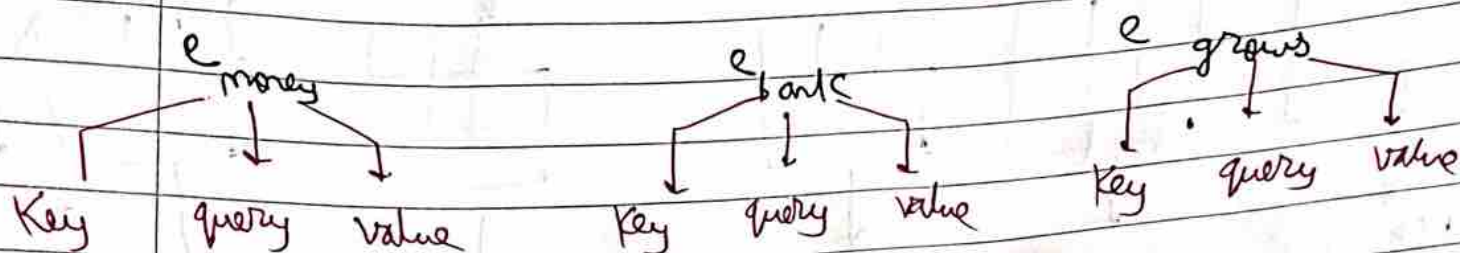


The 1<sup>st</sup> one is called query  $e_{money}$  word embedding vector  
Why? because it asks a query as:- Tum mere se kitna similar ho (when doing dot product at 1<sup>st</sup> phase),  $e_{money}$  sabse se puchta hai kyunse aur  $e_{bank}$  se aur  $e_{grow}$  se ki tum mere se kitne similar ho.

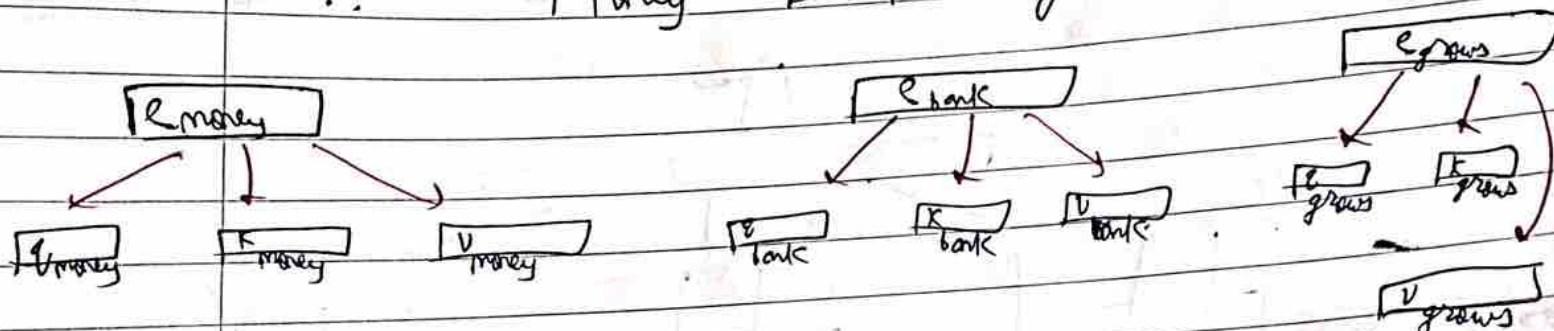
The 2<sup>nd</sup> one is called key  $e_{money}$  word embedding vector  
Why? because it tries to answer proposed by query  $e_{money}$  as:- jab query  $e_{money}$  puchta hai kitne tum kitne similar ho mujhe to () vector parat k jawab deta as a key vector



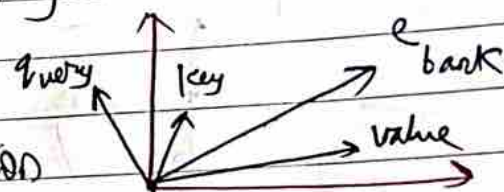
1. The 3<sup>rd</sup> one is called value money word embedding vector because it is the output of query, key vector, hence it is called value money word embedding vector.



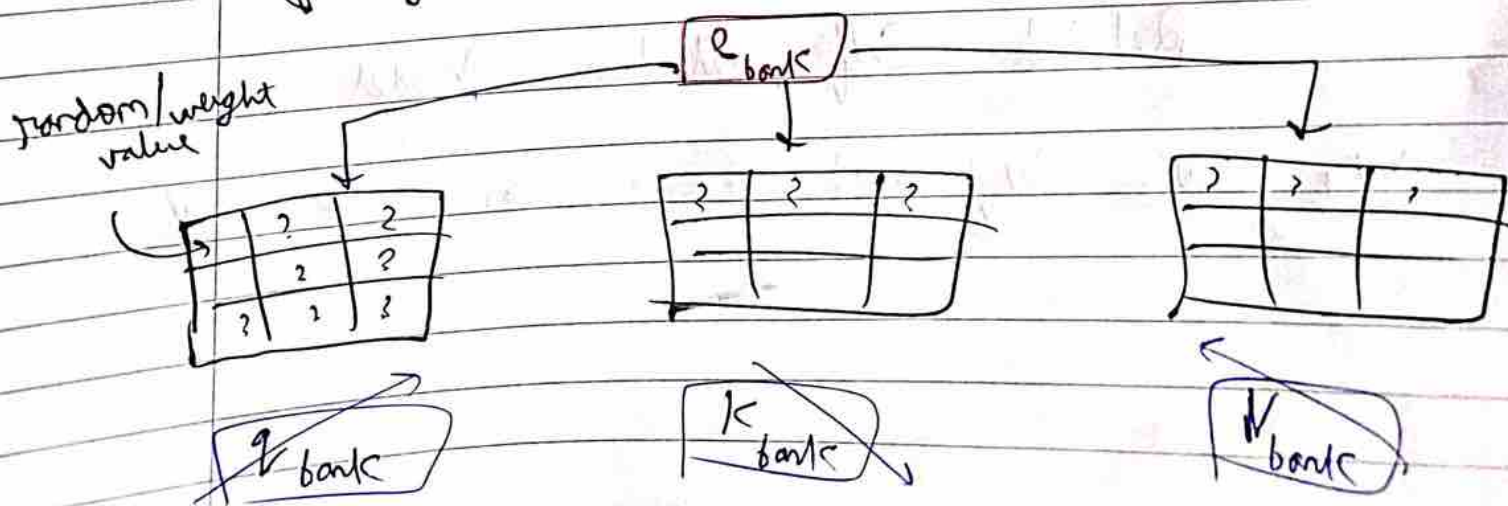
∴ Money Bank grows

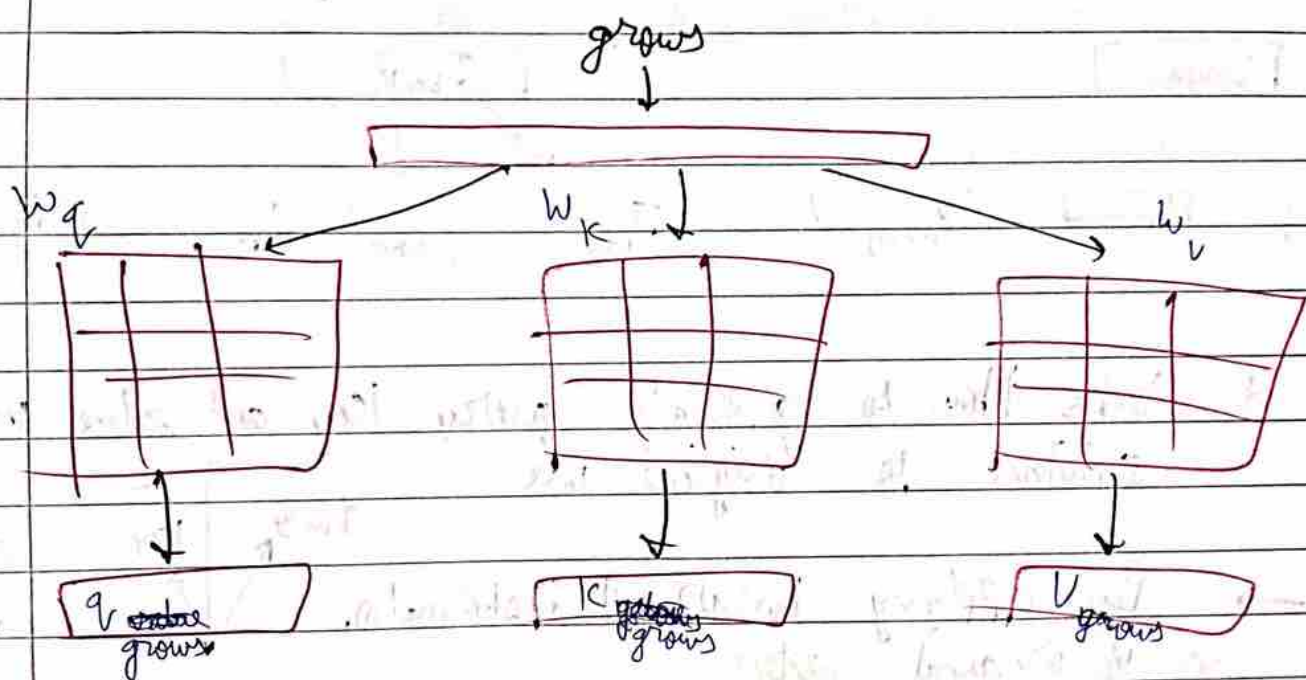
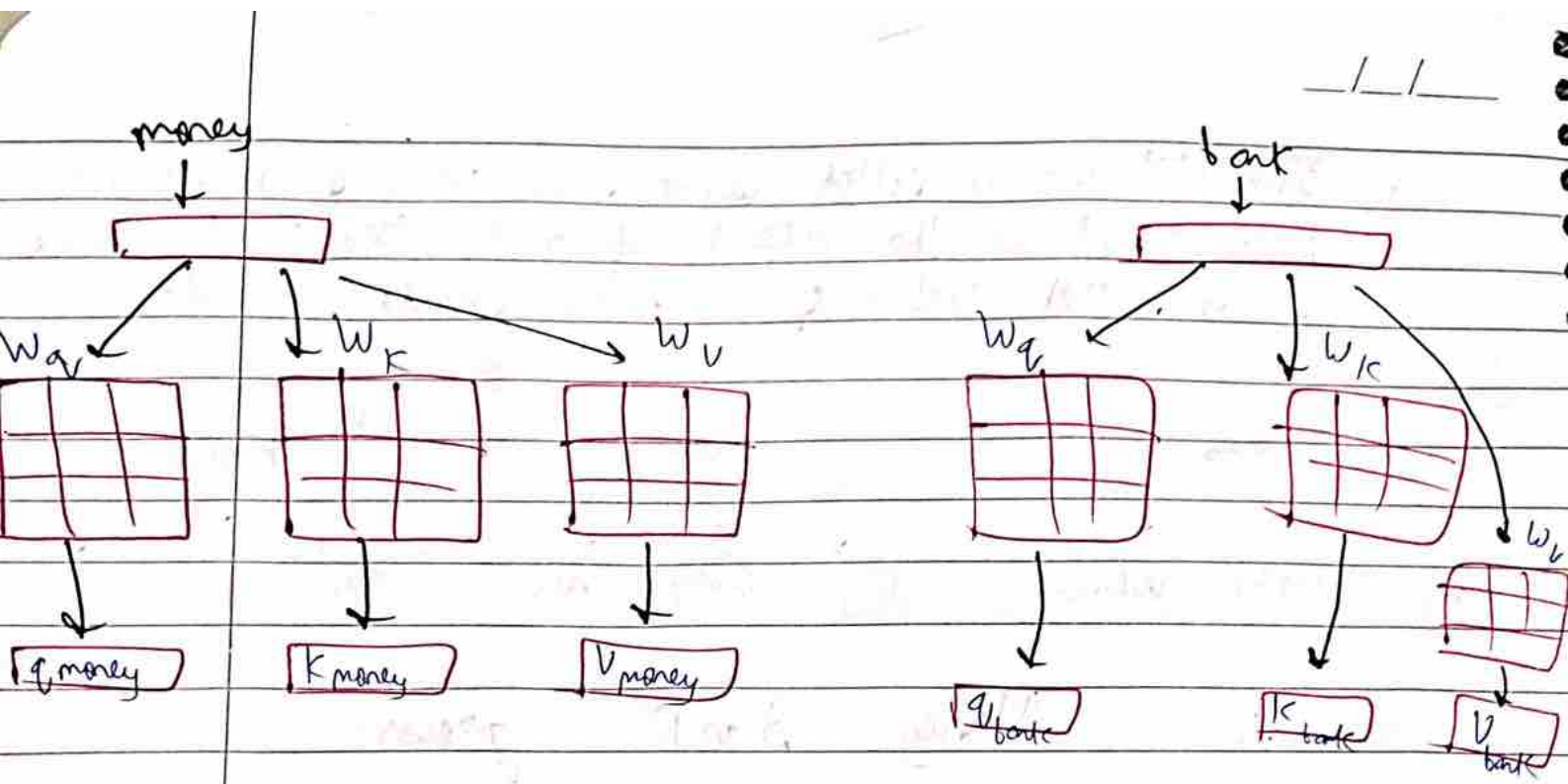


\* ~~How~~ How to generate, query, key and value vectors similar to original one



→ By applying linear transformation on original vector

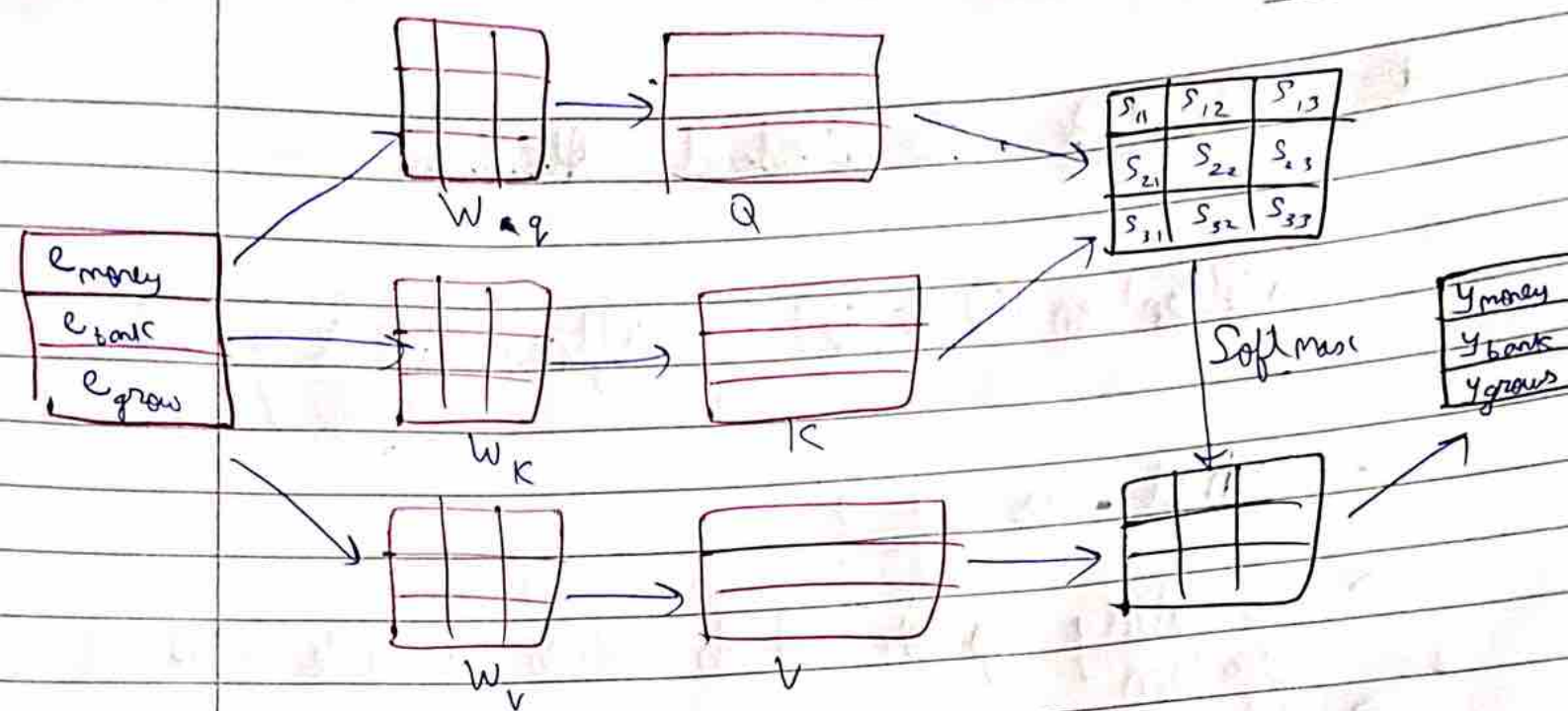




## Refined Self-attention Model

$w_q$ ,  $w_k$ ,  $w_v$  used are all same and equal





Summarized Matrix Attention